

SO(3) Equivariant Neural Networks

Lecture 4 Solutions

From Couplings to a Neural Network

Solutions to checkpoints, discussions, and notebook exercises

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1 Solutions to lecture questions

Question 4.1

A channel-mixing matrix acts on multiplicity indices while $D^{(\ell)}$ acts on magnetic indices, so their actions commute. An arbitrary matrix on m would need to commute with every $D^{(\ell)}(R)$; for an irrep, the only such maps are scalar multiples of the identity. A generic component matrix does not satisfy this condition.

Question 4.2

There are $2 \times 2 = 4$ ordered input-channel pairs for each $1 \otimes 1 \rightarrow L$ path. Each output L has two channels. Therefore the learned path matrix has shape $(2, 4)$ for $L = 0$, $L = 1$, and $L = 2$. The fixed CG tensor shapes differ: $(1, 3, 3)$, $(3, 3, 3)$, and $(5, 3, 3)$.

Question 4.3

A scalar constant is unchanged by rotation, so adding it to an $\ell = 0$ feature is compatible with the transformation law. A fixed nonzero vector b would require $b = Rb$ for every rotation, which is impossible unless $b = 0$. It selects a preferred direction.

Question 4.4

The gate $g(s)$ can be nonlinear because its input s is invariant and its output is a scalar. For a non-scalar multiplet, $y_m = g(s)x_m$ gives $y'_m = g(s)D_{mm'}x_{m'} = D_{mm'}y_{m'}$. The same scalar multiplies every m component.

Question 4.5

No contradiction exists. Equivariance error tests whether the transformation law is obeyed; MSE tests whether the learned equivariant function matches the target. The zero function is exactly equivariant but usually inaccurate.

Question 4.6

More channels add multiplicity within already available geometric types and therefore increase the number of learned combinations. Increasing $\ell_{\max}$ adds new angular types and finer directional structure. The latter usually increases coupling count and computational cost more sharply.

Question 4.7

Fix a nonzero x and let H_x be the set of rotations with $Rx = x$ — the rotations about the axis $\hat{x}$. For any $R \in H_x$, equivariance gives

$$Rf(x) = f(Rx) = f(x), \quad (1)$$

so $f(x)$ is fixed by every rotation about $\hat{x}$. The only vectors in $\mathbb{R}^3$ fixed by all rotations about an axis are the vectors along that axis, so $f(x) = c(x)x$ for some scalar $c(x)$. For $x = 0$, take R to be a rotation by π about each of several axes: $f(0)$ must be fixed by all of them, and those rotations share no common fixed direction, so $f(0) = 0$.

It remains to show c depends only on $\|x\|$. For any rotation R ,

$$c(Rx) Rx = f(Rx) = R f(x) = c(x) Rx, \quad (2)$$

so $c(Rx) = c(x)$: the function c is rotation invariant. Since $SO(3)$ acts transitively on each sphere $\|x\| = r$, any two points of equal norm are related by a rotation, so c is constant on spheres and is therefore a function of $\|x\|$ alone. Writing $g(r)$ for its value on the sphere of radius r gives $f(x) = g(\|x\|)x$.

Two remarks. First, no continuity or linearity was assumed — this is pure orbit bookkeeping, which is why the conclusion is so restrictive. Second, the same argument on a general irrep ℓ is Schur's lemma: the only maps commuting with an irreducible representation are multiples of the identity, so the only freedom left is a scalar, and that scalar must itself be built from invariants. That is exactly the gate.

2 Notebook reference checks

The symmetry-safe pattern is

```
# Learned weights contract channels; CG contracts m indices.
products = np.einsum('Mij,bci,bdj->bcdM', C, x1, x2)
out = np.einsum('op,bpM->boM', W, products.reshape(B, -1, 2*L+1))
```

Three controlled failures are useful:

1. replace one CG tensor by random entries;
2. add a constant vector to an $\ell = 1$ block;
3. apply `np.tanh` separately to vector components.

Each should raise the relative equivariance residual far above floating-point scale. Changing only learned channel weights should not.